

AMAL BAJPAYEE

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🌐 [LinkedIn](#)

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Education

Atal Bihari Vajpayee Indian Institute of Information Technology and Management, Gwalior **2022-2027**
Integrated B.Tech + M.Tech in Information Technology CGPA: 7.05

Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Object Oriented Programming, Machine Learning, Soft Computing

Technical Skills

Languages: C++, Python, JavaScript, SQL

Backend: Node.js, Express.js, FastAPI

Frontend: React.js, HTML, CSS, TypeScript

Databases: MySQL, MongoDB, PostgreSQL

Tools: Git, GitHub, Linux, Postman, VS Code

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Experience

Lord-Tech Datus Solutions Pvt. Ltd.

Gwalior, India

Technical Project Intern (AI & Smart City)

May 2026 – Jul 2026

- Selected as a Core Technical Member for the **TIIC-funded (Rs.2,00,000)** Smart City project **NirmalGwalior**, collaborating directly with the CEO, technical team, and research mentors to build the initial product prototype.
- Designed and developed interactive web and Android application prototypes using AI-assisted development tools, translating product requirements into functional user workflows.
- Participated in system architecture, technology stack selection, and product planning discussions for a scalable AI-enabled municipal waste management platform.
- Coordinated with developers to define implementation strategy, feature roadmap, and technical execution for future production development.

Web Developer Intern

07/2025 – 12/2025

- Developed and customized client-facing web applications and dashboards based on project requirements.
- Debugged backend services and resolved API integration issues to improve application reliability and performance.
- Assisted in maintaining existing websites and implementing responsive user interfaces using modern web technologies.

Projects

NirmalGwalior — AI-Powered Smart City C&D Waste Management Platform

[Web](#) — [App](#)

Rs.2,00,000 TIIC Funded Project — Core Technical Member

- Developed interactive web and mobile application prototypes using AI-assisted development tools to demonstrate the end-to-end Smart City waste management workflow.
- Designed user journeys, UI flows and functional screens for citizen reporting and municipal dashboard operations.
- Collaborated on system architecture, feature planning and technology stack discussions with the engineering team for future production development.
- Contributed to the prototype presented for the TIIC-funded Smart City pilot initiative.

Layer Sequence Extraction of Optimized DNNs Using Side-Channel Information Leaks 05/2025 – 07/2025

Bachelor Thesis Project — Python, PyTorch, Deep Learning, LSTM, Multi-Head Attention

- Investigated side-channel information leakage techniques for architecture inference of optimized deep neural networks.
- Explored sequence modeling approaches using recurrent neural networks and attention mechanisms for reconstructing DNN layer sequences.
- Studied execution-level GPU trace patterns and operator fusion mechanisms to understand their impact on model extraction and architecture reconstruction.

Achievements

- Secured **INR 2,00,000** innovation funding under TIIC, ABV-IIITM Gwalior for the NirmalGwalior Smart City project.
- Successfully completed the **Ethical Hacking and Penetration Testing** program and virtual internship conducted by **C-DAC Noida** under the Cyber Gyan Project.

Certifications

- Deloitte Data Analytics Job Simulation – Forage (Jul 2026)